

Shengxuan Qiu

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Education

Peking University, B.S. in Microelectronics Science and Engineering

Sept 2023 - Jun 2027

School of Electronics Engineering and Computer Science

Beijing, China

- **Major GPA: 3.73/4.00**
- **Incoming Ph.D. student at PKU SEC Lab (Institute for Artificial Intelligence, Peking University)** (starting Fall 2027); research focus on AI infrastructure and AI accelerator architecture.
- **Selected coursework:** Linear Algebra, Machine Learning, Programming in Artificial Intelligence (Operator/Kernel Programming), Probability Theory, Digital Electronics, Analog Electronics, Computer Architecture, Electronic Design Automation (EDA), Semiconductor Physics.

Research Interests

My research focuses on efficient AI systems across the model-system-hardware stack, with an emphasis on scalable reasoning, LLM serving, distributed execution, and hardware-software co-design.

- **AI Infrastructure & LLM Serving:** distributed inference, scheduling, KV-cache management, speculative execution, and serving-system optimization.
- **Reasoning Systems:** test-time scaling, multi-path reasoning, Tree-of-Thought, and agentic workloads.
- **Hardware-Software Co-design:** MoE acceleration, GPU/NPU systems, 3D near-memory processing, and GPU-PIM architectures.

Publications

[1] **HyPER: Bridging Exploration and Exploitation for Scalable LLM Reasoning with Hypothesis Path Expansion and Reduction**

Shengxuan Qiu*, Haochen Huang*, Shuzhang Zhong, Pengfei Zuo, Meng Li

[Paper](#) [Code](#) [Workshop](#) (ICML 2026; *equal contribution)

[2] **Breaking the Reward Barrier: Accelerating Tree-of-Thought Reasoning via Speculative Exploration**

Shuzhang Zhong, Haochen Huang, Shengxuan Qiu, Pengfei Zuo, Runsheng Wang, Meng Li

[Paper](#) [Code](#) (OSDI 2026)

[3] **HDA-MoE: Hybrid Parallelism and Dynamic, Adaptive Scheduling for Mixture-of-Experts with 3D Near-Memory Processing**

Haochen Huang, Shuzhang Zhong, Shengxuan Qiu, Zhe Zhang, Shuangchen Li, Cong Li, Dimin Niu, Hongzhong Zheng, Guangyu Sun, Runsheng Wang, Meng Li

[Paper](#) [Code](#) (IEEE TCAD, 2026)

[4] **S²-MoE: Enabling Efficient Self-Speculative Decoding for Mixture-of-Experts on Edge Devices**

Haochen Huang, Shengxuan Qiu, Meng Li

[Paper](#) [Code](#) (arXiv preprint)

[5] **Tetris: Accelerating Composable Test-Time Scaling on GPU-PIM Systems via Adaptive Mapping, Control, and Memory Management**

Shengxuan Qiu, Haochen Huang, Tenghao Zhao, Meng Li

(HPCA 2027, under review)

Research Experience

PKU SEC Lab, Peking University

Sept 2025 - present

Supervisor: Prof. Meng Li

- **HyPER - dynamic test-time scaling.** Designed and implemented a closed-loop controller for multi-path LLM reasoning that dynamically reallocates compute between exploration and exploitation under a fixed budget; achieved an 8-10% accuracy improvement while reducing token usage by 25-40%.

- **SPEX - speculative Tree-of-Thought serving.** Co-developed an SGLang-based system that breaks reward-dependency barriers via speculative branch expansion, cross-query budget coordination, and adaptive early termination; achieved up to 3x ToT speedup and 4.1x cumulative speedup with token-level speculative decoding.
- **MASBench - multi-agent LLM serving benchmark.** Developing composable multi-agent workloads and causal execution traces for studying serving bottlenecks such as synchronization barriers, context movement, tool stalls, queueing, TTFT/TPOT, and KV-cache behavior across GPU and Ascend platforms.
- **MoE and hardware-software co-design.** Worked on hybrid parallelism and dynamic scheduling for 3D near-memory MoE systems (HDA-MoE), self-speculative MoE inference on edge GPUs (S²-MoE), and adaptive mapping/control for composable test-time scaling on GPU-PIM systems (Tetris).

Selected Technical Experience

- **LLM inference runtime:** modified vLLM execution paths for streaming generation, branching, request abortion, confidence logging, and runtime RoE control; implemented step-level control and MoE routing aggregation.
- **CUDA / GPU programming:** C++/CUDA kernel development, profiling, GPU memory hierarchy, parallel primitives, and performance analysis.
- **Distributed systems:** data/tensor/pipeline/sequence/expert parallelism, MoE communication, scheduling, serving metrics, and workload-level trace/replay.
- **Hardware design:** Verilog/FPGA development, digital-system design, and architecture-level reasoning for AI accelerators and memory-centric systems.

Skills

Programming: Python, C/C++, CUDA, Verilog, Bash, Git, LaTeX

AI Systems: PyTorch, vLLM, SGLang, CUDA kernel programming, distributed inference, MoE, LLM serving and profiling

Mathematics: Good understanding of calculus, linear algebra, probability and statistics, optimization, and numerical methods

Languages: Mandarin (native), English

Achievements

National Scholarship, Top 3%	2025-2026
Peking University Outstanding Student Model (三好学生标兵), Top 2%	2025-2026
Peking University Outstanding Student (三好学生)	2024-2025
Huawei Scholarship	2024-2025